

1. NAME OF THE MEDICINAL PRODUCT

CELCOX 200MG CAPSULES

2. NAME AND STRENGTH OF ACTIVE SUBSTANCES

Celecoxib 200mg

3. PRODUCT DESCRIPTION

CELCOX 200MG CAPSULES

Opaque white/ opaque white Hard gelatin capsules size “1” having imprinting “136” on body with golden yellow ink and “A” on cap with golden yellow ink filled with white to off white colored granular powder.

4. DOSAGE FORM

Capsule

5. PHARMACODYNAMIC PROPERTIES

Pharmacotherapeutic group: Non-steroidal anti-inflammatory and antirheumatic drugs, NSAIDs, Coxibs, ATC code: M01AH01.

Mechanism of action

Celecoxib is an oral, selective, cyclooxygenase-2 (COX-2) inhibitor within the clinical dose range (200-400 mg daily). No statistically significant inhibition of COX-1 (assessed as ex vivo inhibition of thromboxane B₂ [TxB₂] formation) was observed in this dose range in healthy volunteers.

Pharmacodynamic effects

Cyclooxygenase is responsible for generation of prostaglandins. Two isoforms, COX-1 and COX-2, have been identified. COX-2 is the isoform of the enzyme that has been shown to be induced by pro-inflammatory stimuli and has been postulated to be primarily responsible for the synthesis of prostanoid mediators of pain, inflammation, and fever. COX-2 is also involved in ovulation, implantation and closure of the ductus arteriosus, regulation of renal function, and central nervous system functions (fever induction, pain perception and cognitive function). It may also play a role in ulcer healing. COX-2 has been identified in tissue around gastric ulcers in humans but its relevance to ulcer healing has not been established.

The difference in antiplatelet activity between some COX-1 inhibiting NSAIDs and COX-2 selective inhibitors may be of clinical significance in patients at risk of thrombo-embolic reactions. COX-2 selective inhibitors reduce the formation of systemic (and therefore possibly endothelial) prostacyclin without affecting platelet thromboxane.

Celecoxib is a diaryl-substituted pyrazole, chemically similar to other non-arylamine sulfonamides (e.g. thiazides, furosemide) but differs from arylamine sulfonamides (e.g. sulfamethoxazole and other sulfonamide antibiotics).

A dose-dependent effect on TxB₂ formation has been observed after high doses of celecoxib. However, in healthy subjects, in small multiple dose studies with 600 mg BID (three times the highest recommended dose) celecoxib had no effect on platelet aggregation and bleeding time compared to placebo.

Clinical efficacy and safety

Several clinical studies have been performed confirming efficacy and safety in osteoarthritis, rheumatoid arthritis and ankylosing spondylitis. Celecoxib was evaluated for the treatment of the inflammation and pain of osteoarthritis of the knee and hip in approximately 4200 patients in placebo and active controlled trials of up to 12 weeks duration. It was also evaluated for treatment of the inflammation and pain of rheumatoid arthritis in approximately 2100 patients in placebo and active controlled trials of up to 24 weeks duration. Celecoxib at daily doses of 200 mg – 400 mg provided pain relief within 24 hours of dosing. Celecoxib was evaluated for the symptomatic treatment of ankylosing spondylitis in 896 patients in placebo and active controlled trials of up to 12 weeks duration. Celecoxib at doses of 100 mg BID, 200 mg QD, 200 mg BID and 400 mg QD in these studies demonstrated significant improvement in pain, global disease activity and function in ankylosing spondylitis.

Five randomised double-blind controlled studies have been conducted including scheduled upper gastrointestinal endoscopy in approximately 4500 patients free from initial ulceration (celecoxib doses from 50 mg – 400 mg BID). In twelve week endoscopy studies celecoxib (100 – 800 mg per day) was associated with a significantly lower risk of gastroduodenal ulcers compared with naproxen (1000 mg per day) and ibuprofen (2400 mg per day). The data were inconsistent in comparison with diclofenac (150 mg per day). In two of the 12-week studies the percentage of patients with endoscopic gastroduodenal ulceration was not significantly different between placebo and celecoxib 200 mg BID and 400 mg BID.

In a prospective long-term safety outcome study (6 to 15 month duration, CLASS study), 5,800 osteoarthritis and 2,200 rheumatoid arthritis patients received celecoxib 400 mg BID (4-fold and 2-fold the recommended osteoarthritis and rheumatoid arthritis doses, respectively), ibuprofen 800 mg TID or diclofenac 75 mg BID (both at therapeutic doses).

Twenty-two percent of enrolled patients took concomitant low-dose acetylsalicylic acid (≤ 325 mg/day), primarily for CV prophylaxis. For the primary endpoint complicated ulcers (defined as gastrointestinal bleeding, perforation or obstruction) celecoxib was not significantly different than either ibuprofen or diclofenac individually. Also for the combined NSAID group there was no statistically significant difference for complicated ulcers (relative risk 0.77, 95 % CI 0.41-1.46, based on entire study duration). For the combined endpoint, complicated and symptomatic ulcers, the incidence was significantly lower in the celecoxib group compared to the NSAID group, relative risk 0.66, 95% CI 0.45-0.97 but not between celecoxib and diclofenac. Those patients on celecoxib and concomitant low-dose acetylsalicylic acid experienced 4 fold higher rates of complicated ulcers as compared to those on celecoxib alone. The incidence of clinically significant decreases in haemoglobin (>2 g/dL), confirmed by repeat testing, was significantly lower in patients on celecoxib compared to the NSAID group, relative risk 0.29, 95% CI 0.17-

0.48. The significantly lower incidence of this event with celecoxib was maintained with or without acetylsalicylic acid use.

In a prospective randomised 24 week safety study in patients who were aged ≥ 60 years or had a history of gastroduodenal ulcers (users of ASA excluded), the percentages of patients with decreases in haemoglobin (≥ 2 g/dL) and/or haematocrit ($\geq 10\%$) of defined or presumed GI origin were lower in patients treated with celecoxib 200 mg twice daily (N=2238) compared to patients treated with diclofenac SR 75 mg twice daily plus omeprazole 20 mg once daily (N=2246) (0.2% vs. 1.1% for defined GI origin, $p = 0.004$; 0.4% vs. 2.4% for presumed GI origin, $p = 0.0001$). The rates of clinically manifest GI complications such as perforation, obstruction or haemorrhage were very low with no differences between the treatment groups (4-5 per group).

Cardiovascular safety – long-term studies involving subjects with sporadic adenomatous polyps

Two studies involving subjects with sporadic adenomatous polyps were conducted with celecoxib i.e., the APC trial (Adenoma Prevention with Celecoxib) and the PreSAP trial (Prevention of Spontaneous Adenomatous Polyps). In the APC trial, there was a dose-related increase in the composite endpoint of CV death, myocardial infarction, or stroke (adjudicated) with celecoxib compared to placebo over 3 years of treatment. The PreSAP trial did not demonstrate a statistically significant increased risk for the same composite endpoint.

In the APC trial, the relative risks compared to placebo for a composite endpoint (adjudicated) of CV death, myocardial infarction, or stroke were 3.4 (95% CI 1.4 - 8.5) with celecoxib 400 mg twice daily and 2.8 (95% CI 1.1 - 7.2) with celecoxib 200 mg twice daily. Cumulative rates for this composite endpoint over 3 years were 3.0% (20/671 subjects) and 2.5% (17/685 subjects) respectively, compared to 0.9% (6/679 subjects) for placebo. The increases for both celecoxib dose groups versus placebo were mainly due to an increased incidence of myocardial infarction.

In the PreSAP trial, the relative risk compared to placebo for this same composite endpoint (adjudicated) was

1.2 (95% CI 0.6 - 2.4) with celecoxib 400 mg once daily compared to placebo. Cumulative rates for this composite endpoint over 3 years were 2.3% (21/933 subjects) and 1.9% (12/628 subjects), respectively. The incidence of myocardial infarction (adjudicated) was 1.0% (9/933 subjects) with celecoxib 400 mg once daily and 0.6% (4/628 subjects) with placebo.

Data from a third long-term study, ADAPT (The Alzheimer's Disease Anti-inflammatory Prevention Trial), did not show a significantly increased CV risk with celecoxib 200 mg BID compared to placebo. The relative risk compared to placebo for a similar composite endpoint (CV death, myocardial infarction, stroke) was 1.14 (95% CI 0.61 - 2.12) with celecoxib 200 mg twice daily. The incidence of myocardial infarction was 1.1% (8/717 patients) with celecoxib 200 mg twice daily and 1.2% (13/1070 patients) with placebo.

6. PHARMACOKINETIC PROPERTIES

Absorption

Celecoxib is well absorbed reaching peak plasma concentrations after approximately 2-3 hours. Dosing with food (high fat meal) delays absorption of celecoxib by about 1 hour resulting in a T_{max} of about 4 hours and increases bioavailability by about 20%.

In healthy adult volunteers, the overall systemic exposure (AUC) of celecoxib was equivalent when celecoxib was administered as intact capsule or capsule contents sprinkled on applesauce. There were no significant alterations in C_{max} , T_{max} or $T_{1/2}$ after administration of capsule contents on applesauce.

Distribution

Plasma protein binding is about 97 % at therapeutic plasma concentrations and the medicinal product is not preferentially bound to erythrocytes.

Biotransformation

Celecoxib metabolism is primarily mediated via cytochrome P450 2C9. Three metabolites, inactive as COX-1 or COX-2 inhibitors, have been identified in human plasma i.e., a primary alcohol, the corresponding carboxylic acid and its glucuronide conjugate.

Cytochrome P450 2C9 activity is reduced in individuals with genetic polymorphisms that lead to reduced enzyme activity, such as those homozygous for the CYP2C9*3 polymorphism.

In a pharmacokinetic study of celecoxib 200 mg administered once daily in healthy volunteers, genotyped as either CYP2C9*1/*1, CYP2C9*1/*3, or CYP2C9*3/*3, the median C_{max} and AUC₀₋₂₄ of celecoxib on day 7 were approximately 4-fold and 7-fold, respectively, in subjects genotyped as CYP2C9*3/*3 compared to other genotypes. In three separate single dose studies, involving a total of 5 subjects genotyped as CYP2C9*3/*3, single-dose AUC₀₋₂₄ increased by approximately 3-fold compared to normal metabolisers. It is estimated that the frequency of the homozygous *3/*3 genotype is 0.3-1.0% among different ethnic groups.

Patients who are known, or suspected to be CYP2C9 poor metabolisers based on previous history/experience with other CYP2C9 substrates should be administered celecoxib with caution.

No clinically significant differences were found in PK parameters of celecoxib between elderly African-Americans and Caucasians.

The plasma concentration of celecoxib is approximately 100% increase in elderly women (>65 years).

Compared to subjects with normal hepatic function, patients with mild hepatic impairment had a mean increase in C_{max} of 53% and in AUC of 26% of celecoxib. The corresponding values in patients with moderate hepatic impairment were 41% and 146% respectively. The metabolic capacity in patients with mild to moderate impairment was best correlated to their albumin values. Treatment should be initiated at half the recommended dose in patients with moderate liver impairment (with serum albumin 25-35 g/l). Patients with severe hepatic impairment (serum albumin <25 g/l) have not been studied and celecoxib is contraindicated in this patient group.

There is little experience of celecoxib in renal impairment. The pharmacokinetics of celecoxib has not been studied in patients with renal impairment but is unlikely to be markedly changed in these patients. Thus caution is advised when treating patients with renal impairment. Severe renal impairment is contraindicated.

Elimination

Celecoxib is mainly eliminated by metabolism. Less than 1 % of the dose is excreted unchanged in urine. The inter- subject variability in the exposure of celecoxib is about 10-fold. Celecoxib exhibits dose- and time-independent pharmacokinetics in the therapeutic dose range. Elimination half-life is 8-12 hours. Steady state plasma concentrations are reached within 5 days of treatment.

7. INDICATIONS

- For the management of acute pain in adults and for the treatment of primary dysmenorrhea.
- Relief of the acute and chronic pain and inflammation of rheumatoid arthritis and osteoarthritis.
- Relief of signs and symptoms of ankylosing spondylitis.
- For the management of low back pain.

8. RECOMMENDED DOSAGE For Oral Use

Posology and method of administration:

Celecoxib capsules can be taken with or without food.

Given the association between cardiovascular (CV) risk and exposure to COX-2 inhibitors, doctors are advised to use the lowest effective dose for the shortest possible duration of treatment.

Adults

Symptomatic Treatment of Osteoarthritis (OA): The recommended dose of celecoxib is 200 mg administered as a single dose or as 100 mg twice per day.

Symptomatic Treatment of Rheumatoid Arthritis (RA): The recommended dose of celecoxib is 100 mg or 200 mg twice per day.

Ankylosing Spondylitis (AS): The recommended dose of celecoxib is 200 mg administered as a single dose or 100 mg twice per day. Some patients may benefit from a total daily dose of 400 mg.

Management of Acute Pain: The recommended dose of celecoxib is 400 mg, initially, followed by an additional 200 mg dose, if needed on the first day. On subsequent days, the recommended dose is 200 mg twice daily, as needed.

Treatment of Primary Dysmenorrhea: The recommended dose of celecoxib is 400 mg, initially, followed by an additional 200 mg dose, if needed on the first day. On subsequent days, the recommended dose is 200 mg twice daily, as needed.

Low Back Pain (LBP): Usual dosage for adults is 100 mg of celecoxib orally twice daily, morning and evening, or 200 mg once daily (100 mg and 200 mg only).

Elderly: No dosage adjustment is generally necessary. However, for elderly patients with a lower than average body weight (<50 kg), it is advisable to initiate therapy at the lowest recommended dose.

Hepatic Impairment: No dosage adjustment is necessary in patients with mild hepatic impairment (Child- Pugh Class A). Introduce celecoxib at half the recommended dose in arthritis or pain patients with moderate hepatic impairment (Child-Pugh Class B). Patients with severe hepatic impairment (Child-Pugh Class C) have not been studied.

Renal Impairment: No dosage adjustment is necessary in patients with mild or moderate renal impairment. There is no clinical experience in patients with severe renal impairment.

Children:

Celecoxib is not indicated for use in children.

CYP2C9 Poor Metabolizers: Patients who are known, or suspected to be CYP2C9 poor metabolizers based on genotyping or previous history/experience with other CYP2C9 substrates should be administered celecoxib with caution as the risk of dose-dependent adverse effects is increased. Consider reducing the dose to half the lowest recommended dose.

9. CONTRAINDICATIONS

Celecoxib is contraindicated in:

- Patients with known hypersensitivity to celecoxib or any other ingredient of the product.
- Patients with known sulfonamide hypersensitivity.
- Patients who have experienced asthma, urticaria or allergic-type reactions after taking acetylsalicylic acid (ASA [aspirin]) or other non-steroidal anti-inflammatory drugs (NSAIDs), including other cyclooxygenase-2 (COX-2) specific inhibitors.
- Treatment of peri-operative pain in the setting of coronary artery bypass graft (CABG) surgery see Section Special warnings and special precautions for use).
- Patients who have increased risk of cardiovascular disease (ischaemic heart disease and stroke).

10. SPECIAL WARNINGS AND PRECAUTIONS FOR USE

Risk of GI Ulceration, Bleeding and Perforation with NSAID

Serious GI toxicity such as bleeding, ulceration and perforation can occur at any time, with or without warning symptoms, in patients treated with NSAID therapy. Although minor upper GI problems (e.g. dyspepsia) are common, usually developing early in therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms.

Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious GI events and other risk factors associated with peptic ulcer disease (e.g. alcoholism, smoking, and corticosteroid therapy) are at increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events.

Warning to prescriber when prescribing COX-2 Inhibitors to patients with risk factors of heart disease, hypertension (high blood pressure), hyperlipidemia, diabetes, smoking patient and patient with peripheral arterial disease.

Cardiovascular Effects

Cardiovascular Thrombotic Events: Celecoxib may cause an increased risk of serious CV thrombotic events, myocardial infarction (MI), and stroke, which can be fatal.

All NSAIDs may have a similar risk. This risk may increase with dose and duration of use. The relative increase of this risk appears to be similar in those with or without known CV disease or CV risk factors. However, patients with CV disease or CV risk factors may be at greater risk in terms of absolute incidence, due to their increased rate at baseline. To minimize the potential risk for an adverse CV event in patients treated with celecoxib, the lowest effective dose should be used for the shortest duration possible. Physicians and patients should remain alert for the development of such events, even in the absence of previous CV symptoms. Patients should be informed about the signs and symptoms of serious CV toxicity and the steps to take if they occur (see Section Pharmacodynamic Properties).

An increased incidence of myocardial infarction and stroke was found (see Section Contraindications). Celecoxib is not a substitute for acetylsalicylic acid for prophylaxis of CV thromboembolic diseases because of the lack of effect on platelet function. Because celecoxib does not inhibit platelet aggregation, anti-platelet therapies (e.g., acetylsalicylic acid) should not be discontinued.

Hypertension: As with all NSAIDs, celecoxib can lead to the onset of new hypertension or worsening of pre-existing hypertension, either of which may contribute to the increased incidence of CV events.

Patients taking angiotensin converting enzyme (ACE) inhibitors, thiazide diuretics or loop diuretics may have impaired response to these therapies when taking NSAIDs. NSAIDs, including celecoxib, should be used with caution in patients with hypertension. Blood pressure should be monitored closely during the initiation of therapy with celecoxib and throughout the course of therapy.

Fluid Retention and Edema: As with other drugs known to inhibit prostaglandin synthesis, fluid retention and edema have been observed in some patients taking celecoxib. Therefore, patients with pre-existing congestive heart failure (CHF) or hypertension should be closely monitored. Celecoxib should be used with caution in patients with compromised cardiac function, pre-existing edema, or other conditions predisposing to, or worsened by, fluid retention including those taking diuretic treatment or otherwise at risk of hypovolemia.

Gastrointestinal (GI) Effects

Upper and lower GI perforations, ulcers or bleeds have occurred in patients treated with celecoxib. These serious adverse events can occur at any time, with or without warning symptoms, in patients treated with Celecoxib. Patients most at risk of developing these types of GI complications with NSAIDs are the elderly, patients with CV disease, patients using concomitant glucocorticoids, antiplatelet drugs (such as aspirin), or other NSAIDs, patients using alcohol or patients with a prior history of, or active, GI disease, such as ulceration, GI bleeding or inflammatory conditions. Other factors that increase the risk of GI bleeding in patients treated with NSAIDs include longer duration of NSAID therapy; concomitant use of oral corticosteroids, antiplatelet drugs (such as aspirin), anticoagulants; or selective serotonin reuptake inhibitors (SSRIs); smoking; use of alcohol; older age; and poor general health status. Most spontaneous reports of fatal GI events have been in elderly or debilitated patients.

Strategies to Minimize the GI Risks in NSAID-treated Patients:

- Use the lowest effective dosage for the shortest possible duration.
- Avoid administration of more than one NSAID at a time.
- Avoid use in patients at higher risk unless benefits are expected to outweigh the increased risk of bleeding. For such patients, as well as those with active GI bleeding, consider alternate therapies other than NSAIDs.
- Remain alert for signs and symptoms of GI ulceration and bleeding during NSAID therapy.
- If a serious GI adverse event is suspected, promptly initiate evaluation and treatment, and discontinue Celecoxib until a serious GI adverse event is ruled out.
- In the setting of concomitant use of low-dose aspirin for cardiac prophylaxis, monitor patients more closely for evidence of GI bleeding

Renal Effects

NSAIDs including celecoxib may cause renal toxicity. Renal effects are similar to those observed with comparator NSAIDs. Patients at greatest risk for renal toxicity are those with impaired renal function, heart failure, liver dysfunction, and the elderly. Such patients should be carefully monitored while receiving treatment with celecoxib.

Caution should be used when initiating treatment in patients with dehydration. It is advisable to rehydrate patients first and then start therapy with celecoxib.

Advanced Renal Disease

Renal function should be closely monitored in patients with advanced renal disease who are administered celecoxib (see Section Posology and method of administration).

Anaphylactoid Reactions

As with NSAIDs in general, anaphylactoid reactions have occurred in patients exposed to celecoxib (see Section Contraindications).

Serious Skin Reactions

Serious skin reactions, some of them fatal, including drug reaction with eosinophilia and systemic symptoms (DRESS syndrome), exfoliative dermatitis, Stevens-Johnson syndrome, and toxic epidermal necrolysis, have been reported very rarely in association with the use of celecoxib. Patients appear to be at highest risk for these events early in the course of therapy, the onset of the event occurring in the majority of cases within the first month of treatment. Celecoxib should be discontinued at the first appearance of skin rash, mucosal lesions, or any other sign of hypersensitivity.

Hepatic Effects

Patients with severe hepatic impairment (Child-Pugh Class C) have not been studied. The use of celecoxib in patients with severe hepatic impairment is not recommended. Celecoxib should be used with caution when treating patients with moderate hepatic impairment (Child-Pugh Class B), and initiated at half the recommended dose (see Section Posology and method of administration).

Rare cases of severe hepatic reactions, including fulminant hepatitis (some with fatal outcome), liver necrosis, and hepatic failure (some with fatal outcome or requiring liver transplant), have been reported with celecoxib.

A patient with symptoms and/or signs of liver dysfunction, or in whom an abnormal liver function test has occurred, should be monitored carefully for evidence of the development of a more severe hepatic reaction while on therapy with celecoxib.

Use with Oral Anticoagulants

The concomitant use of NSAIDs with oral anticoagulants increases the risk of bleeding and should be given with caution. Oral anticoagulants include warfarin/coumarin-type and novel oral anticoagulants (e.g. apixaban, dabigatran, and rivaroxaban). In patients on concurrent therapy with warfarin or similar agents, serious bleeding events, some of them fatal, have been reported. Because increases in prothrombin time (INR) have been reported, anticoagulation/INR should be

monitored in patients taking a warfarin/coumarin- type anticoagulant after initiating treatment with celecoxib or changing the dose (see Section Interaction with other medicinal products and other forms of interaction).

General

By reducing inflammation, celecoxib may diminish the utility of diagnostic signs, such as fever, in detecting infections.

The concomitant use of celecoxib and non-aspirin NSAID should be avoided. CYP2D6 Inhibition

Celecoxib inhibits CYP2D6. Although it is not a strong inhibitor of this enzyme, a dose reduction may be necessary for individually dose-titrated medicinal products that are metabolised by CYP2D6 (see Section Interaction with other medicinal products and other forms of interaction).

11. INTERACTIONS WITH OTHER MEDICAMENTS

Pharmacodynamic interactions

Anticoagulants

Anticoagulant activity should be monitored particularly in the first few days after initiating or changing the dose of celecoxib in patients receiving warfarin or other anticoagulants since these patients have an increased risk of bleeding complications. Therefore, patients receiving oral anticoagulants should be closely monitored for their prothrombin time INR, particularly in the first few days when therapy with celecoxib is initiated or the dose of celecoxib is changed. Bleeding events in association with increases in prothrombin time have been reported, predominantly in the elderly, in patients receiving celecoxib concurrently with warfarin, some of them fatal.

Anti-hypertensives

NSAIDs may reduce the effect of anti-hypertensive medicinal products including ACE-inhibitors, angiotensin II receptor antagonists, diuretics and beta-blockers. As for NSAIDs, the risk of acute renal insufficiency, which is usually reversible, may be increased in some patients with compromised renal function (e.g. dehydrated patients, patients on diuretics, or elderly patients) when ACE-inhibitors, angiotensin II receptor antagonists, and/or diuretics are combined with NSAIDs, including celecoxib. Therefore, the combination should be administered with caution, especially in the elderly. Patients should be adequately hydrated and consideration should be given to monitoring of renal function after initiation of concomitant therapy, and periodically thereafter.

In a 28-day published clinical study in patients with lisinopril-controlled Stage I and II hypertension, administration of celecoxib 200 mg BID resulted in no clinically significant increases, when compared to placebo treatment, in mean daily systolic or diastolic blood pressure as determined using 24-hour ambulatory blood pressure monitoring. Among patients treated with celecoxib 200 mg BID, 48% were considered unresponsive to lisinopril at the final clinic visit (defined as either cuff diastolic blood pressure >90 mmHg or cuff diastolic blood pressure increased >10% compared to baseline), compared to 27% of patients treated with placebo; this difference was statistically significant.

Ciclosporin and tacrolimus

Co-administration of NSAIDs and ciclosporin or tacrolimus may increase the nephrotoxic effect of ciclosporin or tacrolimus, respectively. Renal function should be monitored when celecoxib and any of these medicinal products are combined.

Acetylsalicylic acid

Celecoxib can be used with low-dose acetylsalicylic acid but is not a substitute for acetylsalicylic acid for CV prophylaxis. In the submitted studies, as with other NSAIDs, an increased risk of gastrointestinal ulceration or other gastrointestinal complications compared to use of celecoxib alone was shown for concomitant administration of low-dose acetylsalicylic acid (see section 5).

Pharmacokinetic interactions

Effects of celecoxib on other medicinal products

CYP2D6 inhibition

Celecoxib is an inhibitor of CYP2D6. The plasma concentrations of medicinal products that are substrates of this enzyme may be increased when celecoxib is used concomitantly. Examples of medicinal products which are metabolised by CYP2D6 are antidepressants (tricyclics and SSRIs), neuroleptics, anti-arrhythmic medicinal products, etc. The dose of individually dose-titrated CYP2D6 substrates may need to be reduced when treatment with celecoxib is initiated or increased if treatment with celecoxib is terminated.

Concomitant administration of celecoxib 200 mg twice daily resulted in 2.6-fold and 1.5-fold increases in plasma concentrations of dextromethorphan and metoprolol (CYP2D6 substrates), respectively. These increases are due to celecoxib inhibition of the CYP2D6 substrate metabolism.

CYP2C19 inhibition

In vitro studies have shown some potential for celecoxib to inhibit CYP2C19 catalysed metabolism. The clinical significance of this *in vitro* finding is unknown. Examples of medicinal products which are metabolised by CYP2C19 are diazepam, citalopram and imipramine.

Methotrexate

In patients with rheumatoid arthritis celecoxib had no statistically significant effect on the pharmacokinetics (plasma or renal clearance) of methotrexate (in rheumatologic doses). However, adequate monitoring for methotrexate-related toxicity should be considered when combining these two medicinal products.

Lithium

In healthy subjects, co-administration of celecoxib 200 mg twice daily with 450 mg twice daily of lithium resulted in a mean increase in C_{max} of 16% and in AUC of 18% of lithium. Therefore, patients on lithium treatment should be closely monitored when celecoxib is introduced or withdrawn.

Oral contraceptives

In an interaction study, celecoxib had no clinically relevant effects on the pharmacokinetics of oral contraceptives (1 mg norethisterone /35 micrograms ethinylestradiol).

Glibenclamide/tolbutamide

Celecoxib does not affect the pharmacokinetics of tolbutamide (CYP2C9 substrate), or glibenclamide to a clinically relevant extent.

Effects of other medicinal products on celecoxib

CYP2C9 poor metabolisers

In individuals who are CYP2C9 poor metabolisers and demonstrate increased systemic exposure to celecoxib, concomitant treatment with CYP2C9 inhibitors such as fluconazole could result in further increases in celecoxib exposure. Such combinations should be avoided in known CYP2C9 poor metabolisers.

CYP2C9 inhibitors and inducers

Since celecoxib is predominantly metabolised by CYP2C9 it should be used at half the recommended dose in patients receiving fluconazole. Concomitant use of 200 mg single dose of celecoxib and 200 mg once daily of fluconazole, a potent CYP2C9 inhibitor, resulted in a mean increase in celecoxib C_{max} of 60% and in AUC of 130%. Concomitant use of inducers of CYP2C9 such as rifampicin, carbamazepine and barbiturates may reduce plasma concentrations of celecoxib.

Ketoconazole and antacids

Ketoconazole or antacids have not been observed to affect the pharmacokinetics of celecoxib.

Paediatric population

Interaction studies have only been performed in adults.

12. FERTILITY, PREGNANCY & LACTATION

Fertility

Based on the mechanism of action, the use of NSAIDs, including celecoxib, may delay or prevent rupture of ovarian follicles, which has been associated with reversible infertility in some women. In women who have difficulties conceiving or who are undergoing investigation of infertility, withdrawal of NSAIDs, including celecoxib, should be considered.

Pregnancy

Celecoxib in animals have shown reproductive toxicity, (see Section Preclinical safety data). The relevance of these data for humans is unknown. Celecoxib, as with other drugs inhibiting prostaglandin synthesis, may cause uterine inertia and premature closure of the ductus arteriosus and should be avoided during the third trimester of pregnancy.

Celecoxib should be used during pregnancy only if the potential benefit to the mother justifies the potential risk to the fetus.

Inhibition of prostaglandin synthesis might adversely affect pregnancy. An increased risk of spontaneous abortion after use of prostaglandin synthesis inhibitors in early pregnancy is suggested. In animals, administration of prostaglandin synthesis inhibitors has been shown to result in increased pre and post-implantation loss.

If used during second or third trimester of pregnancy, NSAIDs may cause fetal renal dysfunction which may result in reduction of amniotic fluid volume or oligohydramnios in severe cases. Such effects may occur shortly after treatment initiation and are usually reversible. Pregnant women on celecoxib should be closely monitored for amniotic fluid volume.

Lactation

In rats celecoxib is excreted in milk at concentrations similar to those in plasma. Administration of celecoxib to lactating women has shown very low transfer of celecoxib into breast milk. Because of the potential for adverse reactions in nursing infants from celecoxib, a decision should be made whether to discontinue nursing or to discontinue the drug, taking into account the expected benefit of the drug to the mother.

13. UNDESIRABLE EFFECTS

The following adverse drug reactions (ADRs) in Table 1 were identified with incidence rates greater than 0.01% in celecoxib group and greater than those reported in placebo group, during treatment duration up to 12 weeks at daily doses from 100 mg up to 800 mg in adults.

The frequencies on the ADRs in Table 1 are updated based on representing clinical exposure in 38,102 patients taking celecoxib. ADR frequencies are defined as: very common (≥10%), common (≥1% and <10%), uncommon (≥0.1% and <1%), rare (≥0.01% and <0.1%) very rare (<0.01%). The ADRs in Table 1 are listed by system organ class and ranked by frequency in descending order.

Table 1. Adverse Drug Reactions (ADRs) frequency from celecoxib used in Daily Doses of 25 mg-800 mg, in Adult Populations

	Common	Uncommon	Rare	Very rare
Infections and infestations	Bronchitis, sinusitis, upper respiratory tract infection, urinary tract infection	Pharyngitis, rhinitis		
Blood and lymphatic system disorders		Anaemia	Thrombocytopenia	
Immune system disorders		Hypersensitivity		

Psychiatric disorders	Insomnia	Anxiety	Confusional state	
Nervous system disorders	Dizziness	Hypertonia, somnolence		
Eye disorders		Vision blurred		
Ear and labyrinth disorders		Tinnitus		
Cardiac disorders		Palpitations	Cardiac failure congestive, arrhythmia, tachycardia	
Vascular disorders	Hypertension (including aggravated hypertension)		Flushing	
Respiratory, thoracic and mediastinal disorders	Cough			
Gastrointestinal disorders	Vomiting, abdominal pain, diarrhoea,	Gastric ulcer, tooth disorder	Duodenal ulcer, oesophageal ulcer	Intestinal perforation, pancreatitis
Hepatobiliary disorders		Hepatic enzyme increased		
Skin and subcutaneous tissue disorders	Pruritus (includes pruritus generalized), rash	Urticaria, ecchymosis	Angioedema, alopecia	Dermatitis bullous
General disorders and administration site conditions	Oedema peripheral edema	Face oedema, influenza-like illness		
Injury, poisoning and procedural conditions		Injury		

The following additional adverse drug reactions in Table 2 were identified with incidence rates greater than placebo.

Frequencies of ADRs in Table 2 were determined and defined as: very common ($\geq 10\%$), common ($\geq 1\%$ and $< 10\%$), uncommon ($\geq 0.1\%$ and $< 1\%$). The ADRs in Table 2 are listed by system organ class and ranked by frequency in descending order.

Table 2. Adverse Reactions from celecoxib Daily Doses of 400 mg–800 mg

	Very Common	Common	Uncommon
Infections and infestations		Ear infection, fungal infection ²	Helicobacter infection, herpes zoster, erysipelas, wound infection, gingivitis, labyrinthitis, bacterial infection

Neoplasms benign, malignant, and unspecified			Lipoma
Psychiatric disorders			Sleep disorder
Nervous system disorders			Cerebral infarction
Eye disorders			Conjunctival hemorrhage, vitreous floaters
Ear and labyrinth disorders			Hypoacusis
Cardiac disorders		Myocardial infarction, angina pectoris	Angina unstable, aortic valve incompetence, arteriosclerosis coronary artery, sinus bradycardia, ventricular hypertrophy
Vascular disorders	Hypertension		Deep vein thrombosis, haematoma
Respiratory, thoracic and mediastinal disorders		Dyspnoea	Dysphonia
Gastrointestinal disorders	Diarrhoea	Vomiting, dysphagia, irritable bowel syndrome, gastroesophageal reflux disease, nausea, diverticulum	Hemorrhoidal haemorrhage, frequent bowel movements, mouth ulceration, stomatitis
Hepatobiliary disorders		Hepatic enzyme increased (includes alanine aminotransferase increased and aspartate aminotransferase increased) ¹	
Skin and subcutaneous tissue disorders			Dermatitis allergic
Musculoskeletal and connective tissue disorders		Muscle spasms	Synovial cyst
Renal and urinary disorders		Nephrolithiasis	Nocturia
Reproductive system and breast disorders		Vaginal haemorrhage, prostatitis, benign prostatic hyperplasia	Ovarian cyst, menopausal symptoms, breast tenderness, dysmenorrhoea

General disorders and administration site conditions			Oedema
Investigations		Blood creatinine increased, prostatic specific antigen increased, Weight increased	Blood potassium increased, blood sodium increased, blood testosterone decreased, haematocrit decreased, haemoglobin increased
Injury, poisoning and procedural complications			Foot fracture, lower limb fracture, fracture, epicondylitis, tendon rupture

¹

Hypertension, vomiting, diarrhoea, and hepatic enzyme increased are included in Table 2 instead of to Table 1

²

Fungal infections were primarily non-systemic.

Post-marketing Experience

Adverse reactions identified from post-marketing experience are provided below. Frequencies are defined as: very common ($\geq 10\%$), common ($\geq 1\%$ and $< 10\%$), uncommon ($\geq 0.1\%$ and $< 1\%$), rare ($\geq 0.01\%$ and $< 0.1\%$), very rare ($< 0.01\%$), not known (cannot be estimated from the available data):

Immune system disorders: Very rare: anaphylactic reaction

Psychiatric disorders: Rare: hallucination

Nervous system disorders: Very rare: cerebral haemorrhage, meningitis aseptic, ageusia, and anosmia

Eye disorders: Uncommon: conjunctivitis

Vascular disorders: Very rare: vasculitis

Respiratory, thoracic and mediastinal disorders: Rare: pulmonary embolism, pneumonitis

Gastrointestinal disorders: Rare: gastrointestinal haemorrhage

Hepatobiliary disorders: Rare: hepatitis; Very rare: hepatic failure, hepatitis fulminant, hepatic necrosis (see Section Special warnings and special precautions for use), cholestasis, hepatitis cholestatic, jaundice

Skin and subcutaneous tissue disorders: Rare: photosensitivity reaction; Very rare: Stevens-Johnson syndrome, erythema multiforme, toxic epidermal necrolysis, drug reaction with eosinophilia and systemic symptoms (DRESS), acute generalised exanthematous pustulosis (AGEP), dermatitis exfoliative

Renal and urinary disorders: Rare: renal failure acute (see Section Special warnings and special precautions for use), hyponatremia; Very rare: tubulointerstitial nephritis, nephrotic syndrome, glomerulonephritis minimal lesion

Reproductive system and breast disorders: Rare: menstrual disorder; Not known: infertility female (female fertility decreased) (see Section Fertility, pregnancy and lactation)³

General disorders and administration site conditions: Uncommon: chest pain

³

Women intending to become pregnant are excluded.

14. OVERDOSAGE AND TREATMENT

There is no clinical experience of overdose. Single doses up to 1200 mg and multiple doses up to 1200 mg twice daily have been administered to healthy subjects for nine days without clinically significant adverse effects. In the event of suspected overdose, appropriate supportive medical care should be provided e.g. by eliminating the gastric contents, clinical supervision and, if necessary, the institution of symptomatic treatment. Dialysis is unlikely to be an efficient method of medicinal product removal due to high protein binding.

15. STORAGE CONDITION

Do not store above 30°C.

Store in the original package in order to protect from moisture.

16. PACKAGING AVAILABLE

1] Blister pack of Aluminium Foil & PVC/ACLAR Film.

Blister pack of 10 capsules

3 such blisters in a carton

10 such blisters in a carton

2] Blister pack of Aluminium Foil & PVC-PE-PVdC Film

Blister pack of 10 capsules

3 such blisters in a carton

10 such blisters in a carton

3] Blister pack of Aluminium Foil & CFB Film

Blister pack of 10 capsules

3 such blisters in a carton

10 such blisters in a carton

17. NAME AND ADDRESS OF MANUFACTURER

Alembic Pharmaceuticals Limited

Formulation Division,

Village-Panelav, P.O. Tajpura, Near Baska, Taluka

Halol Panchmahal, Gujarat State, India.

18. PRODUCT REGISTRATION HOLDER

GENPHARMA SDN. BHD.

Lot 5016, Jalan Teratai,

5 ½ Mile Off Jalan Meru,

41050 Klang, Selangor, Malaysia

19. DATE OF REVISION

21/11/2023

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